SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – APPLICATION BASED QUESTIONS

Course Code:	CSA17	**Course Title:**	Artificial Intelligence
CO Assessed:	CO2	**Assessment:**	Application Based Questions, In-class
Weightage:	_____ 40% of CIA	**Duration / Marks:**	60 mins
CO2 Statement:	Experiment search and game-solving techniques such as Greedy Search, A*, CSP, Minimax, and Alpha-Beta pruning with heuristic functions for solving complex AI problems efficiently. (BL3)		
Student Name:	Prabhakaran	**Reg. No.:**	1912260199
Section / Batch:	Batch 1	**Date:**	20.10.2026

Code of Conduct

I _____,_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Instructions

- Read each question carefully before answering (in-Class Assessment)
- Each question is a real-world application scenario. Identify the AI technique that fits, apply it step by step, and justify every decision.
- Questions are drawn from Informed Search (Greedy, A*, heuristics), Local Search and Online Search, Constraint Satisfaction Problems, and Adversarial Search (Minimax, Alpha-Beta, evaluation functions).
- Show intermediate tables (open lists, domains, α/β values) and state the final answer clearly.
- Marks will be awarded using the Application Based Questions rubric: understanding of problem context, application of concepts, problem-solving approach, accuracy of solution and clarity.
- Upload your work in GCR and Github.

Section / Topic	Focus	Question Nos.
1. Informed Search Strategies	Greedy best-first search, A* search, heuristic functions (admissibility, consistency)	**Q1**
2. Local Search, Online Search & Constraint Satisfaction	Local search and optimization problems, online search agent, CSP (MRV, degree heuristic, forward checking)	**Q2, Q3**
3. Game Playing: Adversarial Search	Games, minimax algorithm, alpha-beta pruning, evaluation functions	**Q4**

Rubrics for Application Based Questions

Criteria	Weightage	Level 4 – Exemplary (100% of marks)	Level 3 – Proficient (75%)	Level 2 – Developing (50%)	Level 1 – Beginning (25%)
Understanding of Problem Context	**20%**	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	**30%**	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	**20%**	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	**20%**	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	**10%**	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Q. No. / Criteria	Understanding of Problem Context (2)	Application of Concepts (3)	Problem-Solving Approach (2)	Accuracy of Solution (2)	Clarity (1)	Sub. Total (10)	Total %
1							
2							
3							
4							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Annexure A – Application Based Questions

Question 1: Campus Delivery Robot – Greedy Best-First vs A*

Topic: Informed search strategies, heuristic functions | 15 minutes

A campus delivery robot must carry a parcel from the Store (S) to the Hostel (G). The one-way road network and the robot's heuristic estimate h(n) of the remaining travel cost to G (in minutes) are given below.

From	To	Cost
S	A	1
S	B	4
A	C	2
A	D	5
B	G	5
C	D	1
D	G	2

Node	S	A	B	C	D	G
h(n)	6	5	3	3	2	0

Assume ties are broken alphabetically and that a node is re-opened if a cheaper path to it is found.

(a) Apply Greedy Best-First Search from S to G. Write the order in which nodes are expanded, the path returned and its total cost.

(b) Apply A* search. Show the open list with g(n), h(n) and f(n) after each expansion, the order of expansion, and the final path and cost.

(c) Check whether h(n) is admissible and consistent by verifying the required condition for every node and every road.

(d) Compare the outputs of (a) and (b). Explain why Greedy Best-First Search returned a different result and which algorithm you would deploy for the robot, with justification.

Question 2: Wi-Fi Access Point Placement – Local Search and Online Search

Topic: Local search algorithms and optimization problems, online search agent | 15 minutes

A technician wants to place a Wi-Fi access point at the best position along a corridor with 10 candidate positions x = 0 to 9. The measured signal quality f(x) (higher is better) is:

x	0	1	2	3	4	5	6	7	8	9
f(x)	2	5	9	6	4	7	12	10	8	3

A move shifts the access point to an adjacent position ($x - 1$ or $x + 1$). Steepest-ascent hill climbing moves to the better neighbour and stops when no neighbour is strictly better.

(a) Run steepest-ascent hill climbing from the start positions x = 0, x = 4 and x = 9. Show each step and the final position and signal quality.

(b) Identify all local maxima and the global maximum. If the start position is chosen uniformly at random from the 10 positions, find the probability that one run reaches the global maximum, the probability that at least one of 3 random-restart runs does, and the expected number of runs needed.

(c) A simulated annealing search is at x = 2 and proposes the move to x = 3. Compute the acceptance probability $e^{(\Delta/T)}$ at T = 2 and at T = 1, and explain what this does to the chance of escaping the local maximum as T falls.

(d) The same search is later done by a robot that cannot see the whole corridor, can only measure f(x) where it stands, and cannot undo exploration for free. Explain why this is an online search problem, and state what the agent must remember to avoid cycling.

Question 3: Exam Timetable Scheduling – Constraint Satisfaction Problem

Topic: Constraint satisfaction problem (MRV, degree heuristic, forward checking) | 15 minutes

Five exams A, B, C, D and E must each be given one slot from {1, 2, 3}. Two exams that share students must not be in the same slot. The pairs sharing students are: A–B, A–C, A–D, B–C, B–D, C–E, D–E. Because of invigilator availability, exam A can only be in slots 1 or 2, and exam E only in slots 1 or 2.

Exam	Initial domain
A	{1, 2}
B	{1, 2, 3}
C	{1, 2, 3}
D	{1, 2, 3}
E	{1, 2}

Rules for the search: choose the variable using Minimum Remaining Values (MRV), break ties with the degree heuristic, then alphabetically; try values in ascending order; use forward checking after every assignment.

(a) Draw the constraint graph (list its edges) and compute the degree of every exam.

(b) Select the first exam to assign using MRV and the degree heuristic, justify the choice, assign its first value, and show the domains of all exams after forward checking.

(c) Continue the search with the same rules, showing the chosen exam, its value and the updated domains at each step, until a complete assignment is found. Verify it against every constraint.

(d) Interpret: explain how MRV, the degree heuristic and forward checking reduced the search effort here, and what forward checking would signal if a domain became empty.

Question 4: Board-Game AI – Minimax, Alpha-Beta Pruning and Evaluation Function

Topic: Game playing, adversarial search, evaluation functions | 15 minutes

A game-playing agent (MAX) searches two plies ahead: MAX chooses one of three moves (B, C, D), then MIN replies with one of three moves. Positions at depth 2 are scored by the evaluation function

$$\mathbf{Eval = 2 \times M + 1 \times Mo}$$

where M is the material advantage of MAX and Mo is the mobility advantage of MAX. The leaf features (M, Mo) are given for the leaves in left-to-right order:

MIN node	Leaf 1 (M, Mo)	Leaf 2 (M, Mo)	Leaf 3 (M, Mo)
B	(1, 1)	(5, 2)	(3, 2)
C	(1, 0)	(1, 2)	(2, 2)
D	(6, 2)	(2, 1)	(0, 2)

(a) Compute the Eval value of all nine leaves.

(b) Run plain minimax. Give the backed-up value of B, C and D, the value at the root and the move MAX should play.

(c) Run alpha-beta pruning on the same tree (left to right). Show α and β at each node, mark every pruned branch, and state the root value and the number of leaves evaluated.

(d) If the leaves under C were ordered (6, 4, 2) instead, how many leaves would alpha-beta evaluate? Explain the effect of move ordering and why pruning never changes the minimax value.

ANSWERS – APPLICATION BASED QUESTIONS

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Each question carries 25 marks. The rubric below follows the criteria and weightage given in the original assessment tool.

Question 1: Campus Delivery Robot – Greedy Best-First vs A*

(a) Greedy Best-First Search

Greedy Best-First Search selects the node with the smallest heuristic h(n).

Step	Expanded node	Frontier after expansion (node: h)
1	S	B:3, A:5
2	B	G:0, A:5
3	G (goal)	A:5

Expansion order: S → B → G.

Returned path: S → B → G. Total cost = 4 + 5 = 9 minutes.

(b) A* Search

A* selects the node with the smallest f(n) = g(n) + h(n).

After expanding	Open list (node: g, h, f)
S	A: (1,5,6); B: (4,3,7)
A	C: (3,3,6); B: (4,3,7); D: (6,2,8)
C	D: (4,2,6); B: (4,3,7) — D updated with cheaper path
D	G: (6,0,6); B: (4,3,7)
G	Goal reached; B remains in the open list

Expansion order: S → A → C → D → G.

Final path: S → A → C → D → G. Total cost = 1 + 2 + 1 + 2 = 6 minutes.

(c) Admissibility and consistency

The true shortest remaining costs are: h*(S)=6, h*(A)=5, h*(B)=5, h*(C)=3, h*(D)=2, h*(G)=0. Every given h(n) is less than or equal to h*(n), so the heuristic is admissible.

Road	Consistency check h(u) ≤ c(u,v)+h(v)	Result
S–A	6 ≤ 1+5 = 6	Pass
S–B	6 ≤ 4+3 = 7	Pass
A–C	5 ≤ 2+3 = 5	Pass
A–D	5 ≤ 5+2 = 7	Pass
B–G	3 ≤ 5+0 = 5	Pass
C–D	3 ≤ 1+2 = 3	Pass
D–G	2 ≤ 2+0 = 2	Pass

All road checks pass; therefore, h(n) is consistent.

(d) Comparison and deployment

Greedy Best-First Search returns a cost of 9 minutes because it considers only h(n) and chooses B because h(B)=3 is smaller than h(A)=5. It does not account for the cost already travelled. A* uses both g(n) and h(n), finds the 6-minute route, and is optimal here. I would deploy A* because it returns the least-cost path when the heuristic is admissible (and this heuristic is also consistent).

Rubric – Question 1 (25 marks)

Criterion	Weight in original rubric	Marks / 25
Understanding of problem context	20%	5
Application of concepts	30%	7.5
Problem-solving approach	20%	5
Accuracy of solution	20%	5
Clarity	10%	2.5
Total	100%	25

Question 2: Wi-Fi Access Point Placement – Local Search and Online Search

(a) Steepest-ascent hill climbing

Start	Steps	Final position and quality
x=0	0(2) → 1(5) → 2(9); neighbours of 2 are 5 and 6, both lower	x=2, f(2)=9
x=4	4(4) → 5(7) → 6(12); neighbour 7 has 10, lower than 12	x=6, f(6)=12
x=9	9(3) → 8(8) → 7(10) → 6(12)	x=6, f(6)=12

The notation x(value) gives the position and its signal quality.

(b) Local/global maxima and random restarts

Local maxima: x=2 with f(2)=9 and x=6 with f(6)=12. The global maximum is x=6 with quality 12.
The starting positions that reach x=6 are x=4, 5, 6, 7, 8 and 9. Thus, P(global maximum in one run)=6/10=0.6.
For three independent random-restart runs, P(at least one reaches the global maximum)=1−(1−0.6)^3=1−0.4^3=0.936 (93.6%).
The expected number of runs to success is 1/p = 1/0.6 = 5/3 ≈ 1.67 runs.

(c) Simulated annealing

The proposed move is x=2 to x=3. Δ = f(3)−f(2)=6−9=−3.
At T=2: acceptance probability = e^(Δ/T)=e^(−3/2) ≈ 0.2231 (22.31%).
At T=1: acceptance probability = e^(−3) ≈ 0.0498 (4.98%).
As temperature falls, a worse move is less likely to be accepted. Therefore, escaping a local maximum by accepting a worse move becomes less likely.

(d) Online search

It is an online search problem because the robot cannot inspect the full corridor in advance and can measure signal quality only at its current position. It must discover the environment while acting, and exploration has a cost. To avoid cycling, it should remember visited positions, observed signal values, and previously tried moves (or maintain a small map/state history).

Rubric – Question 2 (25 marks)

Criterion	Weight in original rubric	Marks / 25
Understanding of problem context	20%	5
Application of concepts	30%	7.5
Problem-solving approach	20%	5
Accuracy of solution	20%	5
Clarity	10%	2.5
Total	100%	25

Question 3: Exam Timetable Scheduling – Constraint Satisfaction Problem

(a) Constraint graph and degrees

Vertices: A, B, C, D and E.

Edges (exams sharing students): A–B, A–C, A–D, B–C, B–D, C–E and D–E.

Exam	Neighbours	Degree
A	B, C, D	3
B	A, C, D	3
C	A, B, E	3
D	A, B, E	3
E	C, D	2

(b) First variable and forward checking

Initial domain sizes: A=2, E=2, and B=C=D=3. MRV gives a tie between A and E. The degree heuristic chooses A because degree(A)=3 is greater than degree(E)=2.

Assign A=1 (the first value in ascending order). Forward checking removes slot 1 from its neighbours B, C and D.

Exam	Domain after A=1
A	{1} (assigned)
B	{2,3}
C	{2,3}
D	{2,3}
E	{1,2}

(c) Continue search

Step	Chosen exam and value	Domains after forward checking
1	A=1	A={1}; B={2,3}; C={2,3}; D={2,3}; E={1,2}
2	B=2	A={1}; B={2}; C={3}; D={3}; E={1,2}
3	C=3	A={1}; B={2}; C={3}; D={3}; E={1,2}
4	D=3	A={1}; B={2}; C={3}; D={3}; E={1,2}
5	E=1	A={1}; B={2}; C={3}; D={3}; E={1}

Complete assignment: A=1, B=2, C=3, D=3, E=1.

Verification: A differs from B, C and D; B differs from C and D; C differs from E; and D differs from E. Every listed constraint is satisfied.

(d) Interpretation

MRV selects an exam with the fewest available slots, so failure is exposed early. The degree heuristic breaks the initial tie by selecting A, which constrains more unassigned exams. Forward checking removes values that conflict with each assignment and detects a dead end immediately if any unassigned exam's domain becomes empty; the algorithm must then backtrack.

Rubric – Question 3 (25 marks)

Criterion	Weight in original rubric	Marks / 25
Understanding of problem context	20%	5
Application of concepts	30%	7.5
Problem-solving approach	20%	5
Accuracy of solution	20%	5
Clarity	10%	2.5
Total	100%	25

Question 4: Board-Game AI – Minimax, Alpha-Beta Pruning and Evaluation Function

(a) Evaluation of all leaves

Eval = 2 × M + Mo.

MIN node	Leaf	(M, Mo)	Calculation	Eval
B	1	(1,1)	2×1+1	3
B	2	(5,2)	2×5+2	12
B	3	(3,2)	2×3+2	8
C	1	(1,0)	2×1+0	2
C	2	(1,2)	2×1+2	4
C	3	(2,2)	2×2+2	6
D	1	(6,2)	2×6+2	14
D	2	(2,1)	2×2+1	5
D	3	(0,2)	2×0+2	2

(b) Plain minimax

MIN chooses the smallest leaf value under each move: B=min(3,12,8)=3; C=min(2,4,6)=2; D=min(14,5,2)=2. MAX chooses max(3,2,2)=3 at the root. Therefore, the root value is 3 and MAX should play move B.

(c) Alpha-beta pruning, left to right

Start at the MAX root with α=−∞ and β=+∞.

Node/leaf visited	Value	α	β	Action
B (MIN), leaf B1	3	−∞	3	β becomes 3
B, leaf B2	12	−∞	3	β remains min(3,12)=3
B, leaf B3	8	−∞	3	β remains 3; B returns 3
Root (MAX)	3	3	+∞	α becomes 3
C (MIN), leaf C1	2	3	2	β=2 ≤ α=3; prune C2 and C3
D (MIN), leaf D1	14	3	14	β=14
D, leaf D2	5	3	5	β=5
D, leaf D3	2	3	2	D returns 2

Pruned branches: C's second and third leaves. Evaluated leaves = 3 under B + 1 under C + 3 under D = 7. Root value remains 3.

(d) New ordering under C: (6, 4, 2)

The root's α is 3 when C is explored. C's first value 6 gives β=6, which is greater than α=3; the second value 4 gives β=4, also greater than α=3. The third value 2 is therefore evaluated and then C returns 2. No leaf under C can be pruned in this order.

Alpha-beta evaluates 3 leaves under B, 3 under C and 3 under D, for a total of 9 leaves. Move ordering affects how much pruning occurs, but not the minimax value, because alpha-beta only skips branches that cannot affect the final MAX/MIN decision.

Rubric – Question 4 (25 marks)

Criterion	Weight in original rubric	Marks / 25
Understanding of problem context	20%	5
Application of concepts	30%	7.5
Problem-solving approach	20%	5
Accuracy of solution	20%	5
Clarity	10%	2.5
Total	100%	25